

SOLVED PRACTICE WORKBOOK

Landforms of Glaciation / Himalayan Glaciers-GLOF - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

Landforms of Glaciation / Himalayan Glaciers-GLOF	
REFRESHED TEACHING NAVIGATION — BASIC/CORE FIRST	
01 FOUNDATION — WHAT MAKES FLOWING GLACIER ICE Persistent snow without flow is not a glacier.	02 FOUNDATION — ACCUMULATION, ABLATION AND ELA Mass-balance system: Accumulation adds snow and refrozen water.
03 FOUNDATION — FLOW, RETREAT AND DEBRIS COVER Evidence chain: Forward flow retreat paradox - Debris-cover	04 FOUNDATION — PLUCKING AND ABRASION Technically, FOUNDATION — Plucking and abrasion is analysed by
05 CORE — CIRQUE-TO-TROUGH EROSIONAL SUITE A U-valley is glacial, a V-valley is primarily fluvial.	06 CORE — FJORDS AND RIAS Evidence chain: Fjord discriminator Qualified use: Prepare the direct
07 CORE — TILL, MORaine, ESKER AND OUTWASH Meltwater deposits: Eskers are sinuous ridges deposited by subglacial	08 CORE — GROUNDED AND FLOATING ICE Evidence chain: Ice-mass sea-level rule Qualified use: Apply the
09 CORE — HIMALAYAN GLACIER MAP HOOKS Evidence chain: Himalayan map discipline Qualified use: Solve	10 CORE — GLACIER LOSS AND WATER TIMING Evidence chain: Water-timing effect Qualified use: Prepare the 2020
11 CORE — HOW GLACIAL LAKES FORM Glacial-lake formation: Retreat can expose overdeepened basins or	12 CORE — GLOF TRIGGER-TO-BREACH CHAIN GLOF is a release mechanism, not a synonym for mountain flood.
13 SYNTHESIS — HAZARD, EXPOSURE AND SOUTH LHONAK South Lhonak boundary: South Lhonak was a moraine-dammed lake	14 SYNTHESIS — MONITORING AND RISK REDUCTION Current evidence boundary: ISRO SAC's Himalayan Glacier Inventory
15 SYNTHESIS — VERIFIED ROUTES AND CRYOSPHERE SPINE Evidence chain: Verified PYQ routes - Current evidence boundary	

Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly explains Glacier definition?

A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. B. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. C. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: A. Explanation: A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. The other options describe different processes, locations, scales or governance categories.

Q2. Which option is the safest spatial interpretation of Glacier definition?

A. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. B. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. C. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. D. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery.

Answer: B. Explanation: A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. The other options describe different processes, locations, scales or governance categories.

Q3. Which statement preserves the process boundary for Glacier definition?

A. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. B. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. C. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. D. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat.

Answer: C. Explanation: A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. The other options describe different processes, locations, scales or governance categories.

Q4. Which option avoids the main UPSC trap concerning Glacier definition?

A. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. D. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow.

Answer: D. Explanation: A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. The other options describe different processes, locations, scales or governance categories.

Q5. Which statement correctly explains Mass-balance system?

A. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. D. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus.

Answer: A. Explanation: Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. The other options describe different processes, locations, scales or governance categories.

Q6. Which option is the safest spatial interpretation of Mass-balance system?

A. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. B. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. C. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. D. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery.

Answer: B. Explanation: Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. The other options describe different processes, locations, scales or governance categories.

Q7. Which statement preserves the process boundary for Mass-balance system?

A. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. D. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.

Answer: C. Explanation: Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. The other options describe different processes, locations, scales or governance categories.

Q8. Which option avoids the main UPSC trap concerning Mass-balance system?

A. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. B. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. C. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: D. Explanation: Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. The other options describe different processes, locations, scales or governance categories.

Q9. Which statement correctly explains ELA, snowline and snout?

A. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. B. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. C. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. D. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery.

Answer: A. Explanation: The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. The other options describe different processes, locations, scales or governance categories.

Q10. Which option is the safest spatial interpretation of ELA, snowline and snout?

A. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. B. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. C. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. D. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms.

Answer: B. Explanation: The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. The other options describe different processes, locations, scales or governance categories.

Q11. Which statement preserves the process boundary for ELA, snowline and snout?

A. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. B. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. C. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. D. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.

Answer: C. Explanation: The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. The other options describe different processes, locations, scales or governance categories.

Q12. Which option avoids the main UPSC trap concerning ELA, snowline and snout?

A. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. B. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus.

Answer: D. Explanation: The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. The other options describe different processes, locations, scales or governance categories.

Q13. Which statement correctly explains Forward-flow retreat paradox?

A. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. D. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms.

Answer: A. Explanation: Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. The other options describe different processes, locations, scales or governance categories.

Q14. Which option is the safest spatial interpretation of Forward-flow retreat paradox?

A. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. B. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. C. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. D. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

Answer: B. Explanation: Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. The other options describe different processes, locations, scales or governance categories.

Q15. Which statement preserves the process boundary for Forward-flow retreat paradox?

A. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. B. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. C. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. D. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.

Answer: C. Explanation: Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. The other options describe different processes, locations, scales or governance categories.

Q16. Which option avoids the main UPSC trap concerning Forward-flow retreat paradox?

A. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. B. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery.

Answer: D. Explanation: Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. The other options describe different processes, locations, scales or governance categories.

Q17. Which statement correctly explains Debris-cover qualification?

A. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. B. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. C. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. D. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

Answer: A. Explanation: Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. The other options describe

different processes, locations, scales or governance categories.

Q18. Which option is the safest spatial interpretation of Debris-cover qualification?

A. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. D. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form.

Answer: B. Explanation: Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. The other options describe different processes, locations, scales or governance categories.

Q19. Which statement preserves the process boundary for Debris-cover qualification?

A. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. B. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. C. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. D. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

Answer: C. Explanation: Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. The other options describe different processes, locations, scales or governance categories.

Q20. Which option avoids the main UPSC trap concerning Debris-cover qualification?

A. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. B. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. C. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. D. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat.

Answer: D. Explanation: Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. The other options describe different processes, locations, scales or governance categories.

Q21. Which statement correctly explains Glacial erosion?

A. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. B. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.

Answer: A. Explanation: Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. The other options describe different processes, locations, scales or governance categories.

Q22. Which option is the safest spatial interpretation of Glacial erosion?

A. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. B. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

Answer: B. Explanation: Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. The other options describe different processes, locations, scales or governance categories.

Q23. Which statement preserves the process boundary for Glacial erosion?

A. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. B. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. C. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. D. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice.

Answer: C. Explanation: Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. The other options describe different processes, locations, scales or governance categories.

Q24. Which option avoids the main UPSC trap concerning Glacial erosion?

A. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. B. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. C. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. D. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms.

Answer: D. Explanation: Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. The other options describe different processes, locations, scales or governance categories.

Q25. Which statement correctly explains Erosional landform suite?

A. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. B. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

Answer: A. Explanation: Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. The other options describe different processes, locations, scales or governance categories.

Q26. Which option is the safest spatial interpretation of Erosional landform suite?

A. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. B. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

Answer: B. Explanation: Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. The other options describe different processes, locations, scales or governance categories.

Q27. Which statement preserves the process boundary for Erosional landform suite?

A. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. B. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. C. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. D. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

Answer: C. Explanation: Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. The other options describe different processes, locations, scales or governance categories.

Q28. Which option avoids the main UPSC trap concerning Erosional landform suite?

A. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. B. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. C. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. D. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.

Answer: D. Explanation: Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. The other options describe different processes, locations, scales or governance categories.

Q29. Which statement correctly explains Fjord discriminator?

A. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. B. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. C. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. D. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form.

Answer: A. Explanation: A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. The other options describe different processes, locations, scales or governance categories.

Q30. Which option is the safest spatial interpretation of Fjord discriminator?

A. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. B. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. C. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. D. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.

Answer: B. Explanation: A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. The other options describe different processes, locations, scales or governance categories.

Q31. Which statement preserves the process boundary for Fjord discriminator?

A. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. B. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. C. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. D. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.

Answer: C. Explanation: A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. The other options describe different processes, locations, scales or governance categories.

Q32. Which option avoids the main UPSC trap concerning Fjord discriminator?

A. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. B. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. C. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. D. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

Answer: D. Explanation: A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. The other options describe different processes, locations, scales or governance categories.

Q33. Which statement correctly explains Till and moraine?

A. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. B. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. C. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. D. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

Answer: A. Explanation: Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. The other options describe different processes, locations, scales or governance categories.

Q34. Which option is the safest spatial interpretation of Till and moraine?

A. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. B. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. C. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. D. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice.

Answer: B. Explanation: Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. The other options describe different processes, locations, scales or governance categories.

Q35. Which statement preserves the process boundary for Till and moraine?

A. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. B. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.

Answer: C. Explanation: Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. The other options describe different processes, locations, scales or governance categories.

Q36. Which option avoids the main UPSC trap concerning Till and moraine?

A. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. B. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. C. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. D. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form.

Answer: D. Explanation: Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. The other options describe different processes, locations, scales or governance categories.

Q37. Which statement correctly explains Meltwater deposits?

A. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. B. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. C. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. D. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion.

Answer: A. Explanation: Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. The other options describe different processes, locations, scales or governance categories.

Q38. Which option is the safest spatial interpretation of Meltwater deposits?

A. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. B. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. C. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. D. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk.

Answer: B. Explanation: Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. The other options describe different processes, locations, scales or governance categories.

Q39. Which statement preserves the process boundary for Meltwater deposits?

A. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. B. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. C. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. D. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk.

Answer: C. Explanation: Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. The other options describe different processes, locations, scales or governance categories.

Q40. Which option avoids the main UPSC trap concerning Meltwater deposits?

A. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. B. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. C. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. D. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

Answer: D. Explanation: Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. The other options describe different processes, locations, scales or governance categories.

Q41. Which statement correctly explains Ice-mass sea-level rule?

A. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. B. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. C. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. D. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.

Answer: A. Explanation: Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. The other options describe different processes, locations, scales or governance categories.

Q42. Which option is the safest spatial interpretation of Ice-mass sea-level rule?

A. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. B. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. C. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. D. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance.

Answer: B. Explanation: Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. The other options describe different processes, locations, scales or governance categories.

Q43. Which statement preserves the process boundary for Ice-mass sea-level rule?

A. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. B. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. C. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. D. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk.

Answer: C. Explanation: Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. The other options describe different processes, locations, scales or governance categories.

Q44. Which option avoids the main UPSC trap concerning Ice-mass sea-level rule?

A. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. B. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. C. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. D. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice.

Answer: D. Explanation: Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. The other options describe different processes, locations, scales or governance categories.

Q45. Which statement correctly explains Himalayan map discipline?

A. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. B. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. C. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. D. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion.

Answer: A. Explanation: Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. The other options describe different processes, locations, scales or governance categories.

Q46. Which option is the safest spatial interpretation of Himalayan map discipline?

A. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. B. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. C. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. D. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity.

Answer: B. Explanation: Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. The other options describe different processes, locations, scales or governance categories.

Q47. Which statement preserves the process boundary for Himalayan map discipline?

A. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. B. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. C. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. D. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

Answer: C. Explanation: Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. The other options describe different processes, locations, scales or governance categories.

Q48. Which option avoids the main UPSC trap concerning Himalayan map discipline?

A. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. B. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. C. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. D. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.

Answer: D. Explanation: Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. The other options describe different processes, locations, scales or governance categories.

Q49. Which statement correctly explains Water-timing effect?

A. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. B. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. C. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. D. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance.

Answer: A. Explanation: Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. The other options describe different processes, locations, scales or governance categories.

Q50. Which option is the safest spatial interpretation of Water-timing effect?

A. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. B. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. C. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. D. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity.

Answer: B. Explanation: Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. The other options describe different processes, locations, scales or governance categories.

Q51. Which statement preserves the process boundary for Water-timing effect?

A. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. B. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. C. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. D. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

Answer: C. Explanation: Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. The other options describe different processes, locations, scales or governance categories.

Q52. Which option avoids the main UPSC trap concerning Water-timing effect?

A. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. B. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. C. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. D. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion.

Answer: D. Explanation: Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. The other options describe different processes, locations, scales or governance categories.

Q53. Which statement correctly explains Glacial-lake formation?

A. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. B. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. C. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. D. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

Answer: A. Explanation: Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. The other options describe different processes, locations, scales or governance categories.

Q54. Which option is the safest spatial interpretation of Glacial-lake formation?

A. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. B. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. C. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. D. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity.

Answer: B. Explanation: Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. The other options describe different processes, locations, scales or governance categories.

Q55. Which statement preserves the process boundary for Glacial-lake formation?

A. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. B. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. C. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. D. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching.

Answer: C. Explanation: Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. The other options describe different processes, locations, scales or governance categories.

Q56. Which option avoids the main UPSC trap concerning Glacial-lake formation?

A. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. C. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. D. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk.

Answer: D. Explanation: Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. The other options describe different processes, locations, scales or governance categories.

Q57. Which statement correctly explains GLOF mechanism?

A. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. B. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. C. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. D. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

Answer: A. Explanation: A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. The other options describe different processes, locations, scales or governance categories.

Q58. Which option is the safest spatial interpretation of GLOF mechanism?

A. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. B. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. C. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. D. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

Answer: B. Explanation: A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. The other options describe different processes, locations, scales or governance categories.

Q59. Which statement preserves the process boundary for GLOF mechanism?

A. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. B. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. C. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. D. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast.

Answer: C. Explanation: A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. The other options describe different processes, locations, scales or governance categories.

Q60. Which option avoids the main UPSC trap concerning GLOF mechanism?

A. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. B. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. C. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. D. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance.

Answer: D. Explanation: A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. The other options describe different processes, locations, scales or governance categories.

Q61. Which statement correctly explains Hazard-risk distinction?

A. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. B. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. C. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. D. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design.

Answer: A. Explanation: GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. The other options describe different processes, locations, scales or governance categories.

Q62. Which option is the safest spatial interpretation of Hazard-risk distinction?

A. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. B. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. C. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. D. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching.

Answer: B. Explanation: GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. The other options describe different processes, locations, scales or governance categories.

Q63. Which statement preserves the process boundary for Hazard-risk distinction?

A. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. C. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. D. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow.

Answer: C. Explanation: GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. The other options describe different processes, locations, scales or governance categories.

Q64. Which option avoids the main UPSC trap concerning Hazard-risk distinction?

A. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. C. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. D. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity.

Answer: D. Explanation: GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. The other options describe different processes, locations, scales or governance categories.

Q65. Which statement correctly explains South Lhonak boundary?

A. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. B. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. C. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. D. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design.

Answer: A. Explanation: South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. The other options describe different processes, locations, scales or governance categories.

Q66. Which option is the safest spatial interpretation of South Lhonak boundary?

A. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. B. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. C. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. D. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow.

Answer: B. Explanation: South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. The other options describe different processes, locations, scales or governance categories.

Q67. Which statement preserves the process boundary for South Lhonak boundary?

A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. C. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: C. Explanation: South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. The other options describe different processes, locations, scales or governance categories.

Q68. Which option avoids the main UPSC trap concerning South Lhonak boundary?

A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. B. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. C. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. D. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

Answer: D. Explanation: South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. The other options describe different processes, locations, scales or governance categories.

Q69. Which statement correctly explains Risk-reduction stack?

A. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. B. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. C. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. D. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast.

Answer: A. Explanation: Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. The other options describe different processes, locations, scales or governance categories.

Q70. Which option is the safest spatial interpretation of Risk-reduction stack?

A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. B. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. C. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. D. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast.

Answer: B. Explanation: Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. The other options describe different processes, locations, scales or governance categories.

Q71. Which statement preserves the process boundary for Risk-reduction stack?

A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. B. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. C. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: C. Explanation: Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. The other options describe different processes, locations, scales or governance categories.

Q72. Which option avoids the main UPSC trap concerning Risk-reduction stack?

A. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. B. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. C. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. D. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design.

Answer: D. Explanation: Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. The other options describe different processes, locations, scales or governance categories.

Q73. Which statement correctly explains Verified PYQ routes?

A. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. C. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: A. Explanation: Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. The other options describe different processes, locations, scales or governance categories.

Demand decoding: The directive **answer** requires a direct position on “Q73. Which statement correctly explains Verified PYQ routes?”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in “Q73. Which statement correctly explains Verified PYQ routes?”.

Analytical body:

1. **Claim and named evidence:** Q73. Which statement correctly explains Verified PYQ routes? **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** C. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "Q73. Which statement correctly explains Verified PYQ routes?".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Q73. Which statement correctly explains Verified PYQ routes?", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

Q74. Which option is the safest spatial interpretation of Verified PYQ routes?

A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. B. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. C. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: B. Explanation: Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. The other options describe different processes, locations, scales or governance categories.

Demand decoding: Treat "Q74. Which option is the safest spatial interpretation of Verified PYQ routes?" as a process, location, scale, terminology and status problem. Verify every statement independently.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in “Q74. Which option is the safest spatial interpretation of Verified PYQ routes?”.

Analytical body:

1. **Claim and named evidence:** Q74. Which option is the safest spatial interpretation of Verified PYQ routes?
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** B. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** C. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in “Q74. Which option is the safest spatial interpretation of Verified PYQ routes?”.

Executable exam-length answer / compression plan: Fix the phenomenon and spatial setting; trace the controlling mechanism; test direction, sequence, threshold and India/world anchor; eliminate the closest term, map or causation distractor.

Why this earns marks: It preserves exact process-to-pattern reasoning and prevents a familiar place-name or correlation from substituting for causation.

How to improve this answer: For “Q74. Which option is the safest spatial interpretation of Verified PYQ routes?”, state why the nearest distractor fails on process, direction, scale, location, terminology, date or official status.

Q75. Which statement preserves the process boundary for Verified PYQ routes?

A. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. B. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. C. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: C. Explanation: Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. The other options describe different processes, locations, scales or governance categories.

Demand decoding: The directive **answer** requires a direct position on “Q75. Which statement preserves the process boundary for Verified PYQ routes?”, all clauses and scales, a process chain, spatial reconstruction, named India/world

evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in “Q75. Which statement preserves the process boundary for Verified PYQ routes?”.

Analytical body:

1. **Claim and named evidence:** Q75. Which statement preserves the process boundary for Verified PYQ routes?
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** B. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** C. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in “Q75. Which statement preserves the process boundary for Verified PYQ routes?”.

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For “Q75. Which statement preserves the process boundary for Verified PYQ routes?”, replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?

A. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. D. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching.

Answer: D. Explanation: Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. The other options describe different processes, locations, scales or governance categories.

Q77. Which statement correctly explains Current evidence boundary?

A. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. B. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. C. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. D. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow.

Answer: A. Explanation: ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. The other options describe different processes, locations, scales or governance categories.

Q78. Which option is the safest spatial interpretation of Current evidence boundary?

A. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. C. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. D. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus.

Answer: B. Explanation: ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. The other options describe different processes, locations, scales or governance categories.

Q79. Which statement preserves the process boundary for Current evidence boundary?

A. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. D. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery.

Answer: C. Explanation: ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. The other options describe different processes, locations, scales or governance categories.

Q80. Which option avoids the main UPSC trap concerning Current evidence boundary?

A. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. B. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. C. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. D. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast.

Answer: D. Explanation: ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. The other options describe different processes, locations, scales or governance categories.

PYQS AND ANSWER PRACTICE

Demand decoding: Treat “Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?” as a process, location, scale, terminology and status problem. Verify every statement independently.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in “Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?”.

Analytical body:

1. **Claim and named evidence:** Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** C. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** D. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in “Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?”.

Executable exam-length answer / compression plan: Fix the phenomenon and spatial setting; trace the controlling mechanism; test direction, sequence, threshold and India/world anchor; eliminate the closest term, map or causation distractor.

Why this earns marks: It preserves exact process-to-pattern reasoning and prevents a familiar place-name or correlation from substituting for causation.

How to improve this answer: For “Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?”, state why the nearest distractor fails on process, direction, scale, location, terminology, date or official status.

VERIFIED PYQ OWNERSHIP AUDIT

Verified routing retains the 2020 and 2023 GS-I questions and the 2019 glacier-river objective route. Older objective keys unavailable locally remain explicitly unverified.

OWNER PYQ LEDGER EXTRACTS

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md, _PYQ-ROUTING-PRELIMS-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2019, 2023
- **Paper(s):** GS-I, Prelims GS-I
- **Routed question demands:** 2

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2019	Prelims GS-I	38	Himalayan rivers and associated glacier sources matching	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2023	GS-I	6	Formation of fjords and their picturesque character	How are they formed and Why · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- Himalayan rivers and associated glacier sources matching
- Formation of fjords and their picturesque character

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md.

- **Years represented:** 2020
- **Paper(s):** GS-I
- **Routed question demands:** 1

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	GS-I	6	Melting Himalayan glaciers and India's water resources	How will it impact · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- Melting Himalayan glaciers and India's water resources

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

PYQ DEMAND CARD 1 - 2020 GS-I

Demand: How will the melting of Himalayan glaciers have a far-reaching impact on the water resources of India? (150 words)

Status: Verified routed direct Mains demand.

Model solution: Explain phased water timing: local melt may initially increase, but reduced stored ice can later weaken lean-season buffering. Link irrigation, hydropower, sediment, ecology and hazard, then qualify with monsoon rain, snow, groundwater, debris and basin-specific glacier fractions.

Demand decoding: The directive **answer** requires a direct position on “PYQ DEMAND CARD 1 - 2020 GS-I”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in “PYQ DEMAND CARD 1 - 2020 GS-I”.

Analytical body:

1. **Claim and named evidence:** Demand: How will the melting of Himalayan glaciers have a far-reaching impact on the water resources of India? (150 words) **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in “PYQ DEMAND CARD 1 - 2020 GS-I”.

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For “PYQ DEMAND CARD 1 - 2020 GS-I”, replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

PYQ DEMAND CARD 2 - 2023 GS-I

Demand: How are fjords formed and why do they constitute picturesque areas? (150 words)

Status: Verified routed direct Mains demand.

Model solution: Trace valley-glacier plucking and abrasion to an overdeepened U-trough, then deglaciation and marine inundation. Steep walls, deep water and hanging-valley waterfalls explain scenery. Distinguish a ria as a drowned river valley.

Demand decoding: The directive **answer** requires a direct position on “PYQ DEMAND CARD 2 - 2023 GS-I”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in “PYQ DEMAND CARD 2 - 2023 GS-I”.

Analytical body:

1. **Claim and named evidence:** Demand: How are fjords formed and why do they constitute picturesque areas? (150 words) **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect

the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in “PYQ DEMAND CARD 2 - 2023 GS-I”.

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For “PYQ DEMAND CARD 2 - 2023 GS-I”, replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

PYQ DEMAND CARD 3 - 2019 Prelims GS-I

Demand: Matching Himalayan glaciers with associated river systems.

Status: Verified routed objective demand; official key unavailable locally.

Model solution: Use the secure pairs Gangotri-Bhagirathi, Zemu-Teesta, Bara Shigri-Chenab and Siachen-Nubra. Preserve the official options and label any answer inference separately.

Demand decoding: Treat “PYQ DEMAND CARD 3 - 2019 Prelims GS-I” as a process, location, scale, terminology and status problem. Verify every statement independently.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in “PYQ DEMAND CARD 3 - 2019 Prelims GS-I”.

Analytical body:

1. **Claim and named evidence:** Demand: Matching Himalayan glaciers with associated river systems. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Status: Verified routed objective demand; official key unavailable locally. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Model solution: Use the secure pairs Gangotri-Bhagirathi, Zemu-Teesta, Bara Shigri-Chenab and Siachen-Nubra. Preserve the official options and label any answer inference separately. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in “PYQ DEMAND CARD 3 - 2019 Prelims GS-I”.

Executable exam-length answer / compression plan: Fix the phenomenon and spatial setting; trace the controlling mechanism; test direction, sequence, threshold and India/world anchor; eliminate the closest term, map or causation distractor.

Why this earns marks: It preserves exact process-to-pattern reasoning and prevents a familiar place-name or correlation from substituting for causation.

How to improve this answer: For “PYQ DEMAND CARD 3 - 2019 Prelims GS-I”, state why the nearest distractor fails on process, direction, scale, location, terminology, date or official status.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain glacier mass balance and the apparent paradox of a retreating yet forward-flowing glacier. Answer in about 150 words.

Model thesis: Mass balance controls geometry, while the terminus retreats when local loss exceeds ice delivery even though ice motion remains downslope.

Claim -> named evidence -> analysis -> qualification:

- A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow.
- Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.
- The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus.
- Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery.
- Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat.

Qualified conclusion: Mass balance controls geometry, while the terminus retreats when local loss exceeds ice delivery even though ice motion remains downslope.

Demand decoding: The directive **explain** requires a direct position on “Explain glacier mass balance and the apparent paradox of a retreating yet forward-flowing...”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Mass balance controls geometry, while the terminus retreats when local loss exceeds ice delivery even though ice motion remains downslope.

Analytical body:

1. **Claim and named evidence:** A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. **Analysis:** Trace the driver ->

mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Mass balance controls geometry, while the terminus retreats when local loss exceeds ice delivery even though ice motion remains downslope.

Executable exam-length answer / compression plan: For a 10-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise three process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For “Explain glacier mass balance and the apparent paradox of a retreating yet forward-flowing...”, replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate the principal erosional and depositional landforms of valley glaciation. Answer in about 150 words.

Model thesis: Agent, process, form and sediment sorting distinguish plucking-abrasion landforms from till and meltwater deposits.

Claim -> named evidence -> analysis -> qualification:

- Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms.
- Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.
- Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form.
- Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

Qualified conclusion: Agent, process, form and sediment sorting distinguish plucking-abrasion landforms from till and meltwater deposits.

Demand decoding: The directive **answer** requires a direct position on “Differentiate the principal erosional and depositional landforms of valley glaciation. Answer...”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Agent, process, form and sediment sorting distinguish plucking-abrasion landforms from till and meltwater deposits.

Analytical body:

1. **Claim and named evidence:** Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

3. **Claim and named evidence:** Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Agent, process, form and sediment sorting distinguish plucking-abrasion landforms from till and meltwater deposits.

Executable exam-length answer / compression plan: For a 10-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise three process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For “Differentiate the principal erosional and depositional landforms of valley glaciation. Answer...”, replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain fjord formation and distinguish fjords from rias. Answer in about 250 words.

Model thesis: Valley-glacier overdeepening followed by marine inundation creates fjords; drowned fluvial valleys create rias.

Claim -> named evidence -> analysis -> qualification:

- Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.
- A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

Qualified conclusion: Valley-glacier overdeepening followed by marine inundation creates fjords; drowned fluvial valleys create rias.

Demand decoding: The directive **explain** requires a direct position on “Explain fjord formation and distinguish fjords from rias. Answer in about 250 words.”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Valley-glacier overdeepening followed by marine inundation creates fjords; drowned fluvial valleys create rias.

Analytical body:

1. **Claim and named evidence:** Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Valley-glacier overdeepening followed by marine inundation creates fjords; drowned fluvial valleys create rias.

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For “Explain fjord formation and distinguish fjords from rias. Answer in about 250 words.”, replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 4 - 15 MARKS

Question: Assess the far-reaching effects of Himalayan glacier loss on India's water resources. Answer in about 250 words.

Model thesis: The strongest effect is altered timing and reliability in glacierised headwaters, qualified by monsoon, snow, groundwater and basin heterogeneity.

Claim -> named evidence -> analysis -> qualification:

- Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.
- Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion.
- ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast.

Qualified conclusion: The strongest effect is altered timing and reliability in glacierised headwaters, qualified by monsoon, snow, groundwater and basin heterogeneity.

Demand decoding: The directive **assess** requires a direct position on “Assess the far-reaching effects of Himalayan glacier loss on India's water resources. Answer...”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The strongest effect is altered timing and reliability in glacierised headwaters, qualified by monsoon, snow, groundwater and basin heterogeneity.

Analytical body:

1. **Claim and named evidence:** Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or

observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The strongest effect is altered timing and reliability in glacierised headwaters, qualified by monsoon, snow, groundwater and basin heterogeneity.

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For “Assess the far-reaching effects of Himalayan glacier loss on India's water resources. Answer...”, replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 5 - 20 MARKS

Question: Analyse GLOF risk as an interaction of cryospheric change, geomorphic instability and valley-floor development. Answer in about 300 words.

Model thesis: Risk arises from lake-dam-slope hazard plus downstream exposure and vulnerability, not from lake growth or warming alone.

Claim -> named evidence -> analysis -> qualification:

- Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk.
- A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance.
- GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity.
- South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.
- Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design.

Qualified conclusion: Risk arises from lake-dam-slope hazard plus downstream exposure and vulnerability, not from lake growth or warming alone.

Demand decoding: The directive **analyse** requires a direct position on “Analyse GLOF risk as an interaction of cryospheric change, geomorphic instability and valley-...”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Risk arises from lake-dam-slope hazard plus downstream exposure and vulnerability, not from lake growth or warming alone.

Analytical body:

1. **Claim and named evidence:** Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or

observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

3. **Claim and named evidence:** GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

4. **Claim and named evidence:** South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

5. **Claim and named evidence:** Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Risk arises from lake-dam-slope hazard plus downstream exposure and vulnerability, not from lake growth or warming alone.

Executable exam-length answer / compression plan: For a 20-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise six to eight process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For “Analyse GLOF risk as an interaction of cryospheric change, geomorphic instability and valley-...”, replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design a Himalayan cryosphere risk-governance strategy integrating observation, warning, land use and infrastructure. Answer in about 300 words.

Model thesis: A complete strategy joins remote inventory to field thresholds, interoperable warning, evacuation, zoning, dam safety and transparent uncertainty.

Claim -> named evidence -> analysis -> qualification:

- GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity.
- Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design.
- ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast.

Qualified conclusion: A complete strategy joins remote inventory to field thresholds, interoperable warning, evacuation, zoning, dam safety and transparent uncertainty.

Demand decoding: The directive **answer** requires a direct position on “Design a Himalayan cryosphere risk-governance strategy integrating observation, warning, land...”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: A complete strategy joins remote inventory to field thresholds, interoperable warning, evacuation, zoning, dam safety and transparent uncertainty.

Analytical body:

1. **Claim and named evidence:** GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: A complete strategy joins remote inventory to field thresholds, interoperable warning, evacuation, zoning, dam safety and transparent uncertainty.

Executable exam-length answer / compression plan: For a 20-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise six to eight process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Design a Himalayan cryosphere risk-governance strategy integrating observation, warning, land...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.